The University Of Azad Jammu & Kashmir, Muzaffarabad



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Lab#1

Q1: Online Ticket Reservation System

Objective

To develop a use case diagram for an online ticket reservation system that shows the interaction between users and the system.

System Description

An online ticket reservation system allows customers to search for available tickets, select a ticket, make a reservation, complete payment, and receive booking confirmation. The administrator manages tickets and reservations.

Actors

1. Customer: Searches for tickets, selects tickets, makes reservations, completes payment, and cancels reservations.
2. Admin: Manages tickets and views reservations.
3. Payment Gateway: Processes customer payments.

Use Cases

1. Register/Login
2. Search Tickets
3. View Available Tickets
4. Select Ticket
5. Reserve Ticket
6. Make Payment
7. View Booking Confirmation

8. Cancel Reservation
9. Manage Tickets
10. View Reservations
11. Process Payment

Conclusion

The use case diagram represents the main functions of the online ticket reservation system and shows how customers, administrators, and the payment gateway interact with it.

Use Case Diagram:

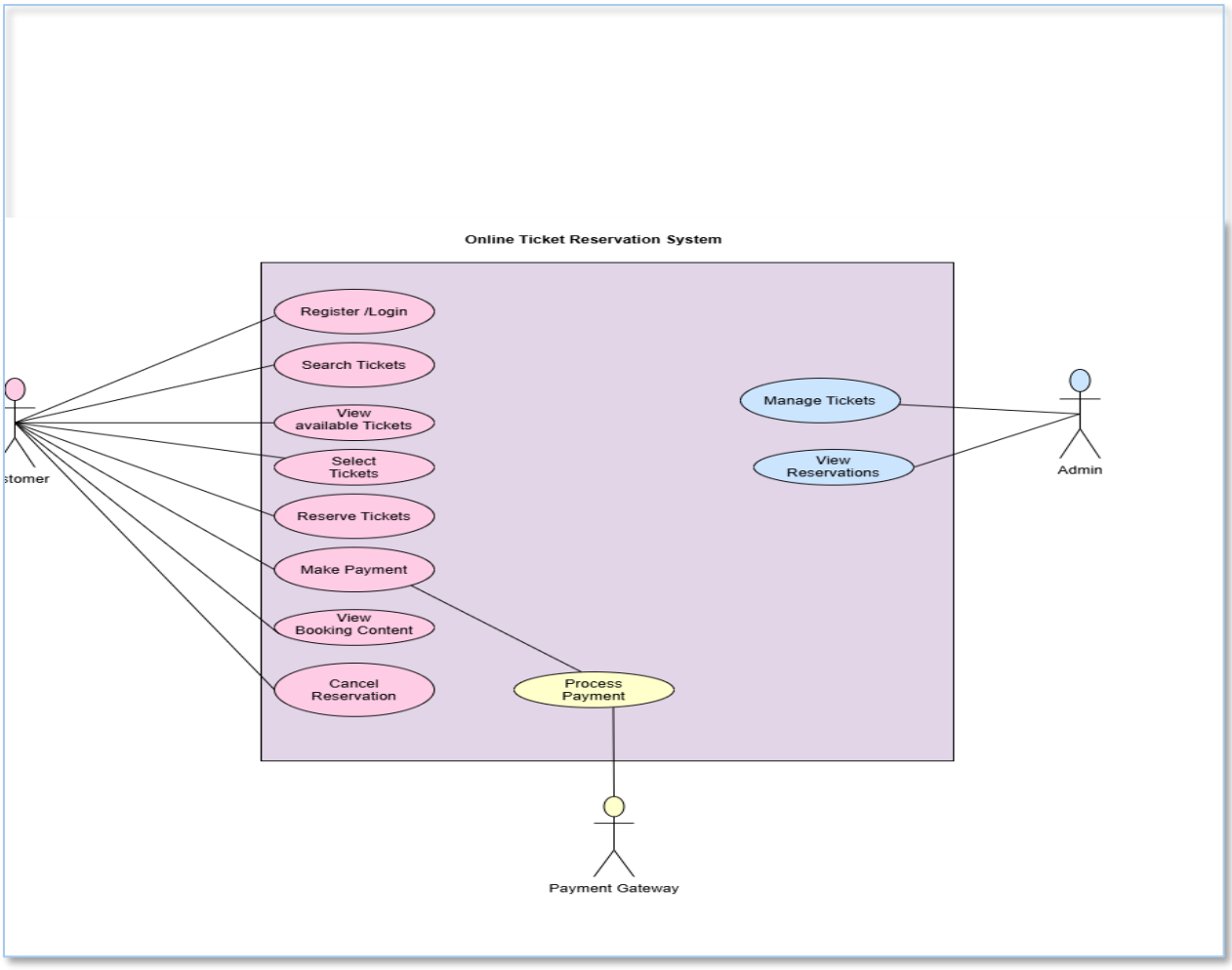


Fig: Use Case Diagram Of Online Reservation System

Q2: Credit Card Processing System

A. Case Study

The Credit Card Processing System is used when a customer purchases an item from a shop using a credit card. The customer selects an item, the vendor prepares a bill, and the customer provides the credit card for processing. The credit card machine processes the transaction and provides a receipt or balance statement. The customer signs the receipt and returns it to the vendor.

B. Identification of Actors

1. Customer

The customer selects an item, provides the credit card for payment, and signs the receipt.

2. Vendor

The vendor prepares the bill, processes the purchase with the credit card machine, and issues the receipt to the customer.

3. Credit Card Machine

The credit card machine reads the credit card, processes the transaction, and generates the receipt or balance statement.

C. Identification of Use Cases

1. Select Item: The customer selects the item to purchase.

2. Generate Bill: The vendor prepares a bill for the selected item.

3. Swipe Credit Card: The customer's credit card is swiped using the card reader.

4. Process Transaction: The credit card machine processes the requested transaction.

5. Generate Receipt: The credit card machine generates the transaction receipt or balance statement.

6. Sign Receipt: The customer signs the receipt.

7. Issue Receipt: The vendor gives the receipt to the customer.

D. Conclusion

The use case diagram represents the interaction between the customer, vendor, and credit card machine during a credit card purchase. It shows the main activities involved in selecting an item, generating a bill, processing payment, and issuing a receipt.

Use Case Diagram

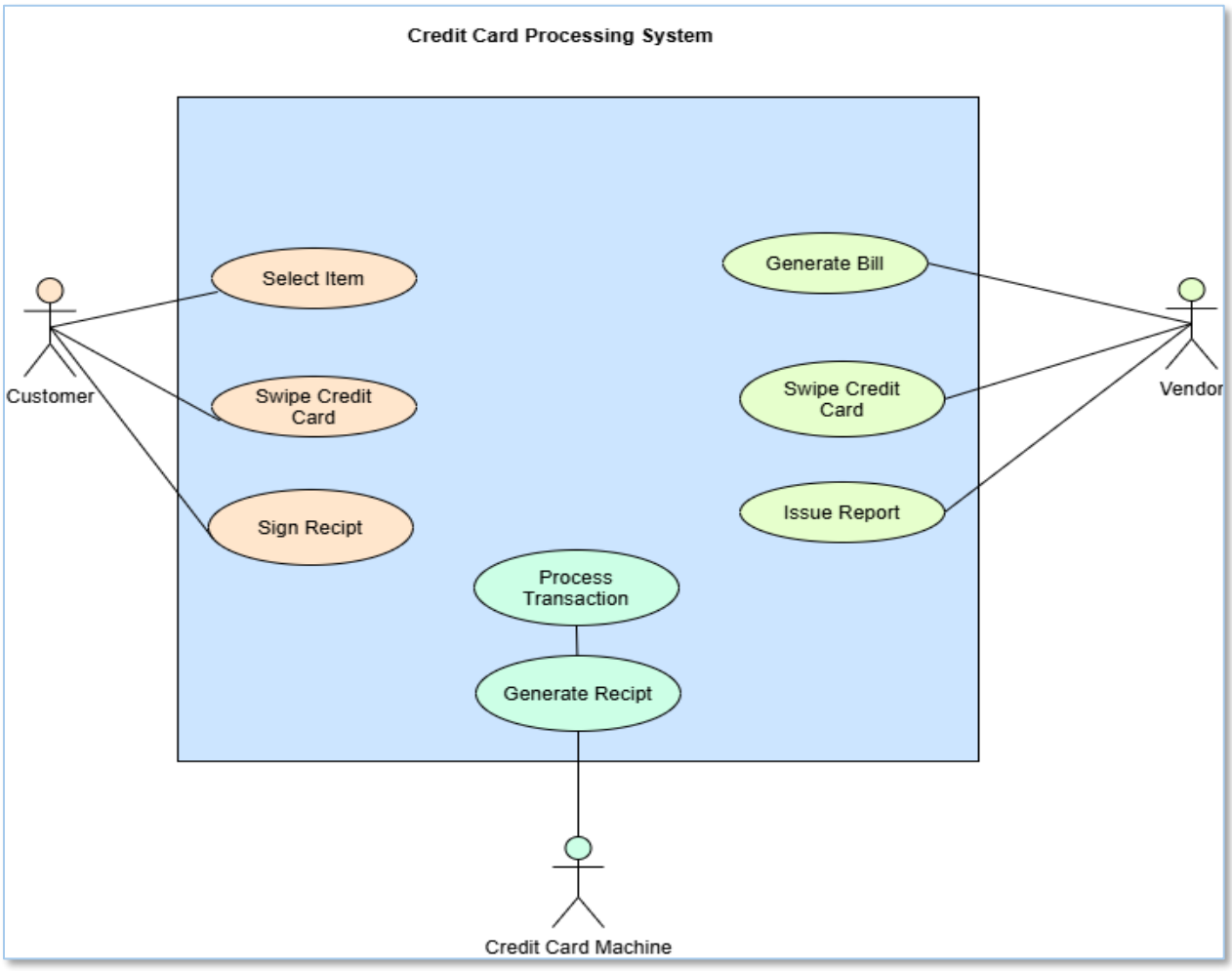


Fig: Use Case Diagram of Credit Card Processing System